

Dachuan Song

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Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University, Fairfax, Virginia, USA

Jan 2024 – Present

College of Engineering and Computing

Ph.D. in Electrical and Computer Engineering, Current GPA: 3.91

Advisor: Prof. Xuan Wang

- **Major Courses:** Linear Systems, Random Processes, Kalman Filtering, Distributed Control and Optimization, System Identification, Advanced Learning from Data

University of Southampton, Southampton, United Kingdom

Sept 2022 – Sept 2023

School of Electronics and Computer Science

M.Sc. in Computer Science, GPA: 3.8 (Top 3%)

- **Major Courses:** Advanced Machine Learning, Advanced Deep Learning, Computer Vision, Bayesian, Active & Reinforcement Learning

Xinjiang University, Xinjiang, China

Sept 2017 – Jun 2021

School of Software

B.Eng. in Software Engineering, GPA: 3.4 (Top 5%)

Publications

CONFERENCE PAPERS AND PREPRINTS

Elastic Spectral State Space Models for Train-Once Budgeted Inference

2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

Accepted to NeurIPS 2026 Main Track (Poster); proceedings forthcoming. arXiv:2601.22488.

Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference

Sep 2026

Dachuan Song, Chuchu Chen, Xuan Wang

arXiv preprint, arXiv:2609.32486, 2026.

Mergeable Model-Side Aggregation States for Long-Context Language Models

Jul 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

arXiv preprint, arXiv:2607.26448, 2026.

Causality-based Subject and Task Fingerprints using fMRI Time-series Data

Nov 2024

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

ACM BCB 2024, Article 13, pp. 1–10 (Oral Presentation). DOI: 10.1145/3698587.3701342.

JOURNAL ARTICLES

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data

Oct 2025

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

Health Information Science and Systems 13, Article 70 (2025). DOI: 10.1007/s13755-025-00388-w.

Research Projects

Elastic Selective Spectral Hybrids for Budgeted Inference

2026 – Present

- Made each spectral channel an input-dependent recurrent unit that can retain or forget earlier information; paired the mixer with sliding-window attention for recent context.
- Trained different channel counts and feed-forward widths together so one checkpoint can export compact models. At 1.53B parameters, B300 batch-one decoding took 1.37 ms/token, $2.14\text{--}2.80\times$ faster than the tested Mamba-2 and Mamba-3 implementations at matched parameter counts.

Elastic Spectral State Space Models for Train-Once Budgeted Inference

Jan 2026 – Present

- Used ordered Hankel spectral channels as an explicit capacity axis: direct prefix truncation removes channels and their weights to export standalone compact models from one training run.
- Trained those prefixes with input-adaptive gates and budget dropout so smaller exports remain useful across inference budgets; evaluated them on language, audio, long-range, and offline reinforcement-learning tasks.

Mergeable Model-Side Aggregation States for Long-Context Language Models

Jul 2026 – Aug 2026

- Kept a 2 KiB mergeable HyperLogLog state alongside frozen language models, carrying count and set evidence across chunks without rereading the full context.
- Reached 99.2% accuracy on 3,969 aggregate-then-reason tasks with Gemma 4 (31B).

Machine Learning for Brain Activity Patterns using EEG (Master's Project)

Apr 2023 – Oct 2023

- Built a CNN-LSTM pipeline for spatial and temporal EEG features, with autoencoder pretraining and attention-based feature learning.
- Applied transfer learning to improve feature extraction from limited EEG data.

Internship Experience

Machine Learning Engineer Intern, Optosurgical

May 2026 – Aug 2026

- Developed video models for surgical phase and instrument-action recognition in robot-assisted procedures, providing workflow-state information for downstream systems.
- Compressed and fine-tuned Depth Anything 3 to 29% of its original size while retaining near-full model performance for monocular depth estimation.

Technical Skills

- **Architectures & Inference:** spectral and selective state space models, recurrent memory, sliding-window attention, long-context processing, elastic model export.
- **ML Engineering:** PyTorch, CUDA, FFT convolution, multi-GPU training, model compression, fast recurrent decoding, reproducible evaluation.
- **Modeling & Control:** linear systems, Kalman filtering, system identification, LQR/MPC, autoregressive models.